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Analytical toxicology

- A sub-discipline of toxicology that provides *tools and techniques* supporting other *toxicology* branches
- Uses analytical chemistry to *assess chemicals* affecting organisms
- Measures *xenobiotics*, including their metabolized forms
- Identifies toxicants by analyzing *biological* samples (e.g., body fluids, stomach contents, solid tissues) or *other materials* (e.g., food, environmental samples).

Application of Analytical toxicology

I. Forensic Applications

- Identifies *chemicals* causing death or injury in humans
- Supports **legal** investigations and proceedings

II. Occupational Toxicology

- Analytical methods also needed for occupation toxicology to monitor worker *exposure* to *toxic* hazards
- Uses *screening* methods to assess threshold limit values
- Ensures workplace *safety*

- **III. Regulatory purposes-**

- Determines if chemicals/drugs pose *low risk* to be *marketed* for a stated purpose
- Sets standards for safe *chemical* levels in *air, water, food*
- Examines *small analyte quantities* in diverse matrices
- Both **occupational and regulatory toxicology** rely on forensic toxicology because an injury or occupational disease in a worker can result in a **legal proceeding**, as well as can occur with a violation of a regulatory law.

- **IV. Experimental studies**

- Measures chemical concentrations (e.g., in blood)
- Supports research on exposure, metabolism, and toxic effects

Forensic Toxicology

- **Origin:** Latin '*forensis*' – public forum (legal disputes settled in public forums)
- **Definition:** Scientific methods applied to legal issues.
- **Forensic toxicology**
 - Uses toxicology in legal settings (*criminal* investigations)
 - Addresses **medico-legal** aspects of harmful chemical effects
 - Establishes *cause-effect* b/n chemical exposure and toxic/lethal outcomes

Medicolegal issues

Three types based on intent: **accidental**, **suicidal**, and **homicidal**.

- **Accidental**- Children drug ingestion, adult overdose, workplace exposure
- **Suicidal**- e.g pesticides (organophosphates), Prescription drugs (like sedatives, antidepressants)
- **Homicidal**- e.g Arsenic, cyanide poison)

Role of forensic toxicologist in postmortem investigations

- Detect & *quantify* drugs/poisons in biological samples in injured person/autopsy.
- *interpret* the analytical findings
- Establish *cause of death* in collaboration with medical examiners & pathologists
- E.g., ethanol in RTA, CO levels in fire victim

The Forensic toxicology laboratory

- Utilize forensic chemistry & modern instruments.
- Analyze blood, urine, tissues for alcohol, drugs, toxins.
- Drugs detected may be either of a *prescribed or illicit nature*(Marijuana, Heroin, LSD, or Cocaine)
- Commonly detected: amphetamines, cocaine, opioids, insecticides, BZD,TCA, cyanide etc
- *licit or illicit psychoactive drugs* often play a significant role in the circumstances associated with sudden or violent death.

Toxicologic Analysis

- If a poison is oral administered
 - Start: GI contents → blood, urine → liver.
- Proper analysis critical to avoid misdiagnosis(case of *limited toxicology analysis*)
- Detect *parent compound or active metabolites (by immunoassays)*
 - (e.g **cocaine**- benzoylecgonine and ecgonine methyl ester).

The toxicologic investigation of a poison death

- ❑ 1. Collect detailed case *history*
- ❑ 2. Gather suitable *specimens* at autopsy:
 - Whole blood, plasma, serum, urine, stomach contents
 - Oral fluid, liver tissue, sweat, *hair*, *nails*, bone marrow, *vitreous* humor, dental
 - *Special cases*:
 - Hair & *nails* for heavy metals and long-term drug use
 - *Maggot* testing in decomposed bodies
- ❑ 3. conducting suitable *toxicologic analyses* based on the available specimens, and
- ❑ 4. the *interpretation* of the analytic findings.

Table 4.3: Viscera and Body Fluids to be Preserved in Suspected Poisoning**1. Routine Viscera & Body Fluids—(to be preserved routinely for analysis, in every case of poisoning)**

<i>Specimen</i>	<i>Quantity</i>
1 Stomach	Entire
2 Stomach contents	Entire (preferably)
3 Small intestine (jejunum)	30 cm (entire length in infants)
4 Small intestinal contents	Up to 100 gm
5 Liver (portion containing gall bladder)	500 gm (entire liver in infants)
6 Kidney	One half of each kidney (both kidneys in infants)
7 Urine	30 – 50 ml
8 Blood	10 – 20 ml

2. Special Viscera—(all or any of these should be preserved in addition to routine viscera, in certain specific cases of poisoning)

<i>Specimen</i>	<i>Quantity</i>	<i>Poison</i>
1 Cerebrospinal fluid	As much as can be withdrawn	Alcohol
2 Brain	Half (one hemisphere)	Alcohol, anaesthetics, barbiturates, carbon monoxide, cyanide, opiates, strychnine
3 Spinal cord	Entire length	Gelsemine, strychnine
4 Lung	One	Inhaled poisons
5 Skin (with underlying tissues)	Affected portion	Corrosives, injected poisons
6 Long bone (preferably femur)	10 cm length	Heavy metals
7 Scalp hair (plucked)	15 to 20 strands	Heavy metals

Histopathological examination

- Microscopic examination of tissue for *organ damage (acute)*
- Evidence of *long-term* abuse which may contributed to *the cause of death*.

Analytical procedures in Forensic toxicology

- Toxicology labs use several methods to screen for poisons/drugs
- ***Unknown substances***: standardized approaches to detect common toxicants
 - ***Gases & volatile substances***: measured by GC
 - ***Nonvolatile organics***: drugs, pesticides, pollutants, analyzed by TLC, HPLC, LC, LC-MS
 - ***Metals***: measured by chemical or thermal oxidation

Analytical procedures in Forensic toxicology...

- *No single* perfect method — combination of screening & confirmatory tests
- Methods *vary* in cost, accuracy, complexity, speed, and specificity
- *Rapid analysis* needed for unstable chemicals before decomposition(CN, CO, ethanol)
- Stable poisons (arsenic, barbiturates, mercury, strychnine) detectable *years later*

i. Sample Collection and Preservation

- Proper *collection*, *storage*, and *preparation* are crucial to avoid contamination or degradation.
- ***Biological*** matrices: blood, urine, vitreous humor, liver tissue, hair, etc.
 - ***Blood:***
- Proper ***labeling***, ***storage*** (refrigeration/freezing), and chain of custody are essential.

ii. Preliminary Screening Tests (Presumptive/Qualitative)

- Used to *rapidly* screen urine samples for drug presence (positive/negative).

❑ Bedside Tests(Color Tests):

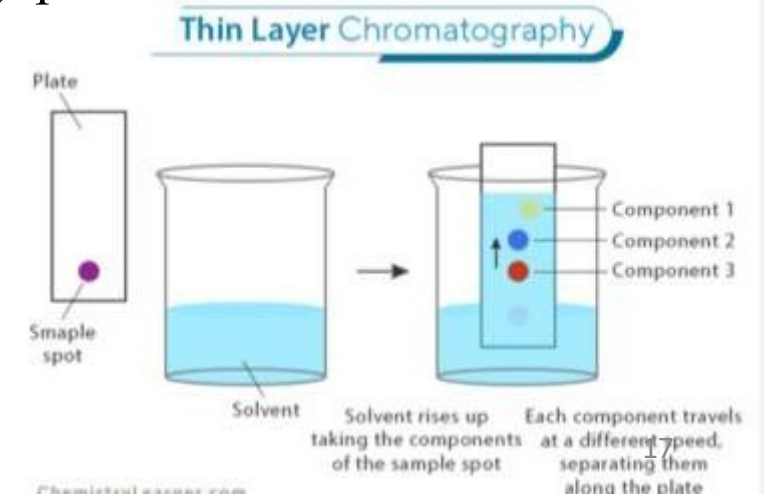
- *Trinder's test*: violet/purple color → salicylates (salicylic acid).
- *Marquis* test: opioids/opiates detection.
- *Scott* test: detects cocaine.
- *Ferric* chloride/perchloric test: purple color → phenol, phenothiazines.

❑ Immunoassays

- Biochemical tests using antigen-antibody binding
- Sensitive to specific drugs/metabolites
- Common methods: **ELISA**, **EMIT**, **Forensic urine drug testing (FUD)**

❑ Thin Layer Chromatography

- involves the movement by capillary action of a *liquid* phase (usually an organic solvent) through a thin, uniform layer of *stationary* phase (usually silica gel) held on a rigid support (usually a glass, aluminum, or plastic sheet).
- Compounds are separated by partition between the *mobile* and *stationary* phases.
- simple, *inexpensive* technique which is widely used.

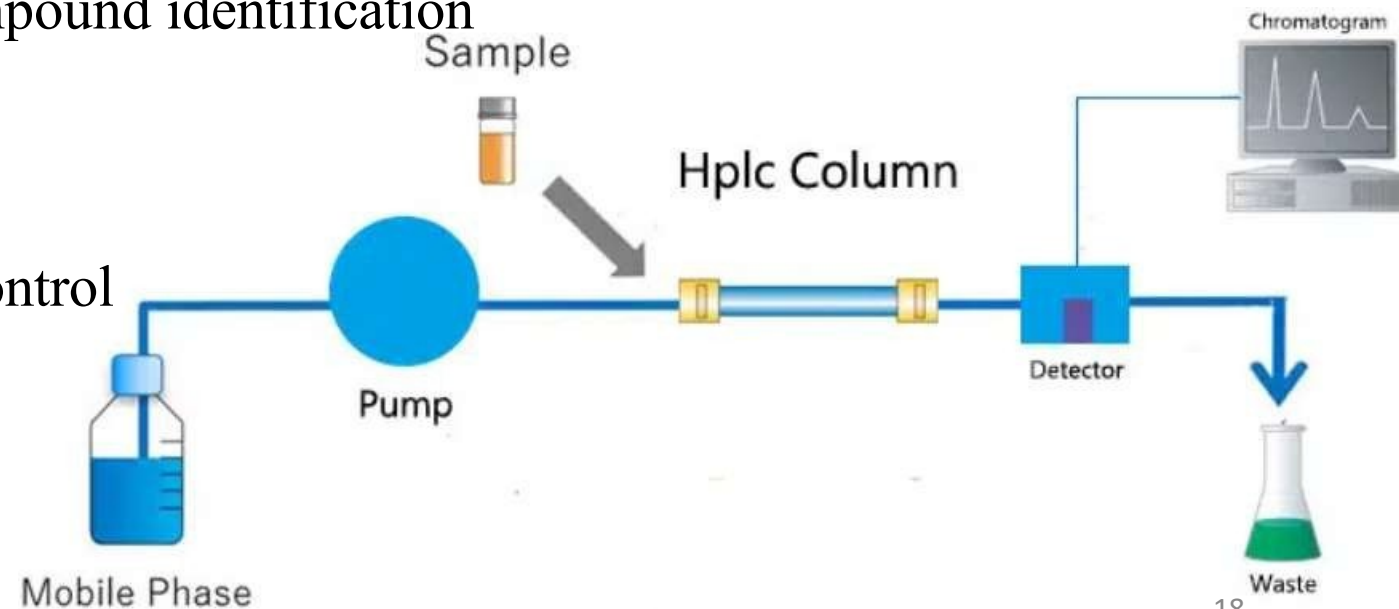


iii. Confirmatory (Quantitative) Analysis

- These methods are highly accurate

1. High-Performance Liquid Chromatography (HPLC)

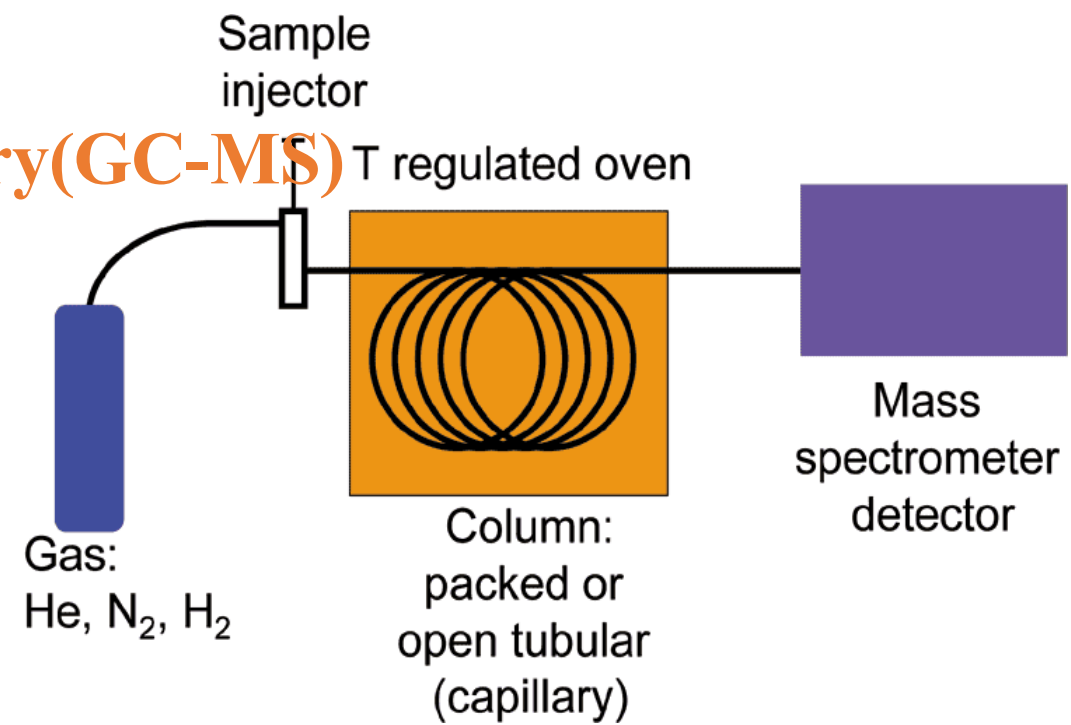
- Separates & quantifies *non-volatile, thermally unstable* compounds:
 - ✓ Alkaloids, flavonoids, peptides, nucleotides, pesticides, most drugs
- **Functions:**
 - ✓ Isolation & purification of compounds
 - ✓ Separates, identifies, quantifies chemicals
- **Detector** types:
 - ✓ UV, fluorescence, refractive index
 - ✓ Often uses reference standards for compound identification
- **Analysis** outputs:
 - ✓ Retention time
 - ✓ Peak area
- Cheaper, widely used for routine quality control



2. Gas Chromatography- volatile cpds

3. Gas Chromatography-Mass Spectrometry(GC-MS)

- ***Sophisticated*** quantitative analysis system
- ***Gold*** standard for drug and poison detection
- ***Mobile*** phase (M.P): *gas*
- ***Stationary*** phase (S.P): *liquid*
- ***Detector***: Mass Spectrometry (MS)
- Separates (GC) and identifies (MS) chemical compounds
- Used for *volatile and semi-volatile* compounds (e.g., alcohol, some drugs & poisons)

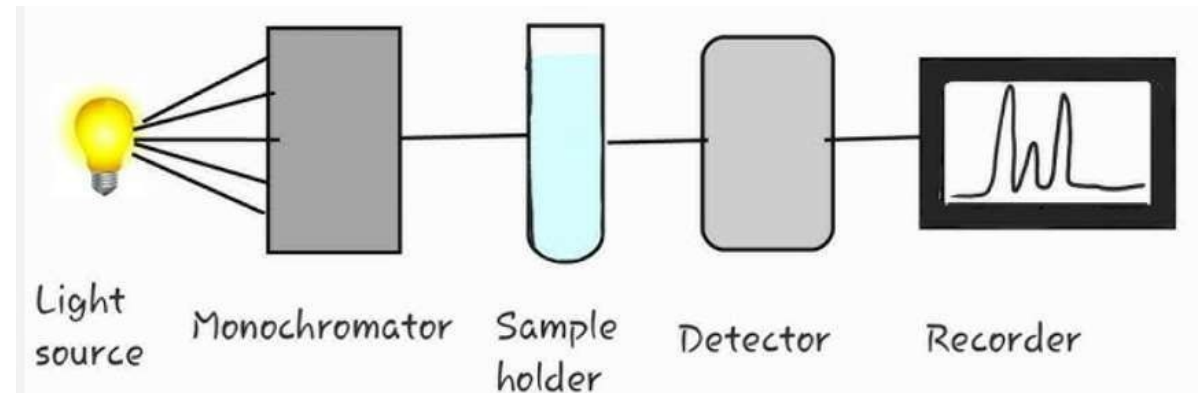


4. Liquid Chromatography-Mass Spectrometry (LC-MS)

- Preferred for *non-volatile and heat-sensitive* compounds
- Used for pharmaceutical *drugs*, toxins, biological metabolites
- ***Detector***: Mass Spectrometry (MS) – mass-based detection
- **GC-MS and LC-MS:**
 - Most widely applied methodologies in toxicology
 - Generally accepted as unequivocal identification of *all drugs*
 - Used in *advanced research and toxicology*

5. Ultraviolet-Visible (UV-Vis) Spectrophotometry 📡

- This technique is based on the principle that many drugs when *in solution* will *absorb UV* radiation (Measures how much UV or visible light a molecule absorbs)
- A *beam of light* from a visible and/or *UV light source* is separated into its component wavelengths by a prism or diffraction grating
- Absorption depends on *chemical structure, concentration, and UVR* wavelength
- Used to analyze organic compounds, *DNA/RNA, proteins, drugs*



6. Atomic Absorption Spectrophotometry

- (AAS) best method for detecting *inorganic elements* (arsenic, lead, mercury, thallium, etc.).
- requires a *large sample* of blood for accurate analysis.
- Measures the absorption of light by free atoms (each element absorbs light at a characteristic wavelength)

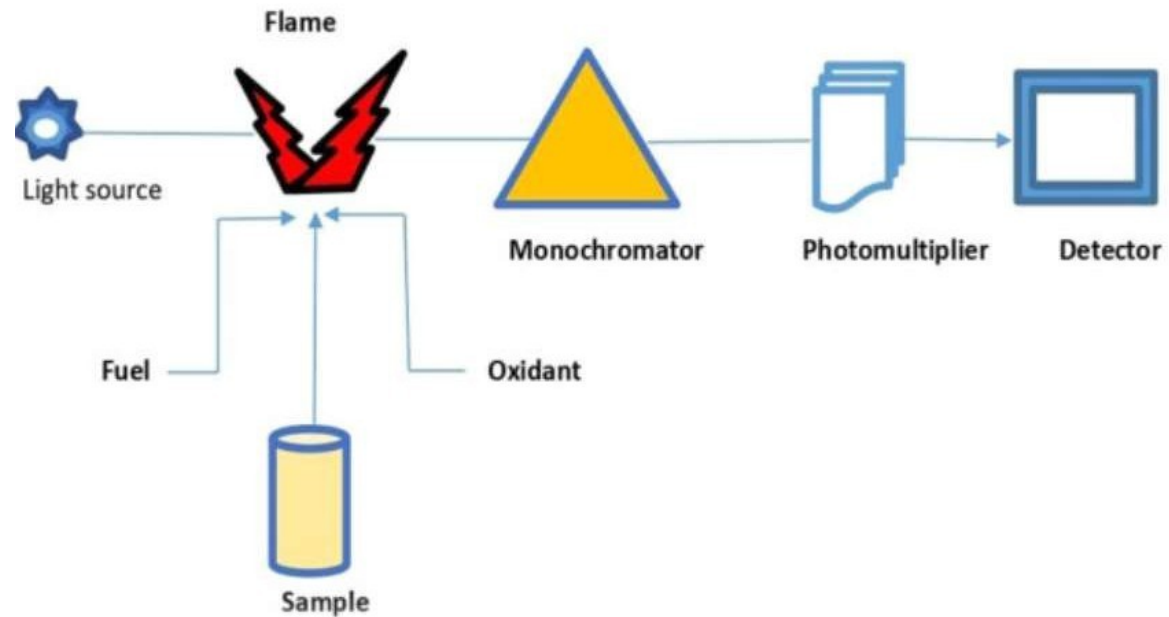


Fig 36.6: Atomic absorption spectrophotometry

Others

❑ Radio Immunoassay (RIA)

- *Slow, expensive*, but highly accurate method for detecting drugs in blood
- Combines ***immunology*** (antibody-antigen binding) with *radioactivity*
- Excellent for detecting drugs at extremely low blood concentrations:
 - Cannabis, digoxin, LSD, paraquat
- *Competitive binding*: radioactively labeled antigen (I-125) vs. sample unlabeled antigen for antibody (measures RA labeled antigen)

❑ Neutron Activation Analysis (NAA)

- Highly *sophisticated* & expensive detection method
- Detects a variety of *inorganic elements* at *ultra-trace* levels
- Sensitivity 100–1000 times better than conventional methods
- Sample bombarded with *neutrons* → forms *radioactive isotopes*
- Isotopes emit *gamma* rays, which are measured and linked to specific elements

iv. Data Interpretation & Reporting

- Analyze *concentration levels* and *pharmacokinetics* (metabolism, redistribution, tolerance, interactions).
- Compare results with *therapeutic*, *toxic*, and *lethal* levels.
- Provide clear, *scientifically* justified *reports*.
- May include *expert testimony* in court.

❖ CRIMINAL POISONING OF LIVING VICTIMS

- Drugs used to *incapacitate* (e.g., cannabinoids, cocaine metabolites, BZDs, amphetamines, opiates, barbiturates, ecstasy).
- *alcohol* is still a primary factor in cases of alleged sexual assault
- Used in *kidnappings*, robberies, *assaults*, child *abuse*, *homicide etc*
- Mechanism- most leads to *sedation* and *incapacitate* the victim and cause amnesia
- ***Delayed*** reporting makes detection difficult.
- Requires sensitive methods like ***GC-MS or LC-MS***.

❖ HUMAN PERFORMANCE TESTING

- Assess presence of ethanol, drugs, and chemicals in specimens.
- Evaluate *their effect* on *human performance* and behavior.
- Commonly applied in **DUI** cases (ethanol, cocaine, marijuana, BZD, opioids).
- Ethanol involved in 28% of motor vehicle accidents (2016).

❖ COURTROOM TESTIMONY

Forensic toxicologists may serve as *expert witnesses*.

- Provide ***objective testimony*** (methods, findings).
- Offer ***opinion testimony*** (*interpretation* of results).

Factors to consider during toxicology analysis

- **Route** of administration (oral, inhalation, parenteral) affects findings.
 - Large amounts of drugs/poisons in the GI tract and liver indicates *oral ingestion*
 - Higher *concentrations in the lungs* than in other visceral organs can indicate *inhalation* route
 - High amount in the tissue surrounding an injection site...*parenteral*
- **Timing** of sample collection is critical (e.g., acetaminophen overdose)
 - *Histopathological* studies may provide additional insights.
- **Urine** results only shows prior exposure, not cause of death

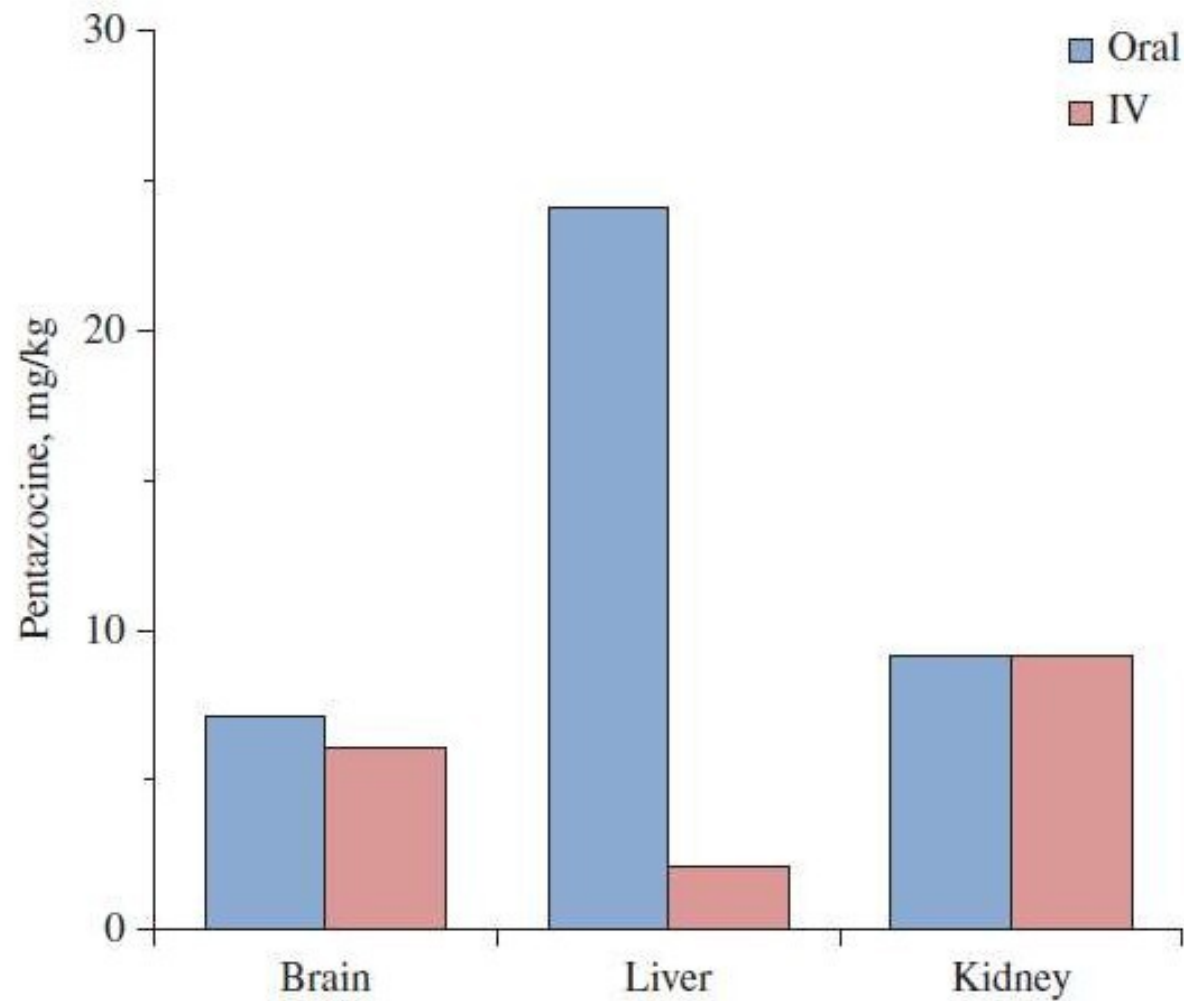


Figure 32-5. Comparison of pentazocine distribution in fatal poisonings due to intravenous injection and oral administration. (Data from Baselt [1982]; Poklis and MacKell [1982].)

❑ Factors to consider during toxicology analysis ...

- Poor drug distribution during *shock* can mislead interpretation
 - e.g., lidocaine - therapeutic level: 5 mg/L, but
 - postmortem blood may exceed 50 mg/L).
- Postmortem redistribution significantly affects interpretation of results
 - Tissue concentrations help distinguish between therapeutic use and overdose
 - Always analyze both *heart* blood, *peripheral* blood, and *tissues*.

Cyanide

- ***Deadly toxic*** substance; *rapid* effects causing death within minutes.
- **Gas:** Hydrogen cyanide (bitter almond smell).
- **Liquid:** Prussic acid, hydrocyanic acid;
 - volatile, flammable.
 - diffuse either by air and explosives
 - Very easy to mix with water
- **Solid:** Sodium, calcium, potassium cyanide (white powder).

Sources

- **Plants:** Cyanogenic glycosides (*cassava*, *bitter* almond, peach, plum).
- **Combustion:** Burning plastics, silk, wool.
- **Cigarette smoke:** 150–200 mcg HCN per cigarette.

Uses of Cyanide

- **Industrial:** Electroplating, metal processing, plastics.
- **Agriculture:** Insecticides, rodenticides.
- **Medicinal:** Sodium nitroprusside (antihypertensive, potential toxicity).
- **Household:** Fumigation, fertilizers, insecticides.

Forensic Issues

- **Accidental:** Industrial, lab, or food-related exposure.
- **Homicide:** Rare; historically used in mass murders (e.g., Nazi genocide).
- **Suicide:** Uncommon, more in medical/lab personnel.
- **Legal** executions (prison use in USA).

Fatal dose and metabolism of CN

- **Fatal dose:** *Hydrocyanic* acid 50–100 mg; *cyanide* salts 100–200 mg.
 - **Bitter almonds:** 50 to 80 almonds

Detoxification Pathways:

1. Main Pathway:

- Rhodanese enzyme (in liver and tissues) converts cyanide (CN^-) to thiocyanate (SCN^-) using sulfur donors (like thiosulfate).
- Thiocyanate is much less toxic and is excreted in urine.

2. Minor Pathways:

- Binding to methemoglobin → cyanmethemoglobins formation.
- Binds with hydroxocobalamin (a form of vit B12 precursor) to form cyanocobalamin (vit B12).
- Small amounts are exhaled as hydrogen cyanide gas/ *bitter almond odor*/

❑ Mechanism of Toxicity

- Inhibits *cytochrome oxidase* → ↓ ATP, ↑ lactic acidosis.
- Inhibits antioxidant enzymes- succinic dehydrogenase, glutathione reductase.
 - Causes direct neurotoxicity.

Clinical Features

- Inhalation produces the ***most rapid*** and serious exposures resulting in almost immediate ***coma***

1. Acute Poisoning:

- ✓ CNS: Headache, anxiety, agitation, confusion, convulsions, dilated pupil and coma.
- ✓ CVS: Initial tachycardia and hypertension, followed by bradycardia and hypotension
- ✓ RS- Tachypnoea followed by bradypnea
- ✓ GIT: Ingestion of cyanide salts leads to nausea, vomiting, and abdominal pain, Acid-base: Anion gap *metabolic acidosis*(lactic acidosis)

2. Chronic Poisoning:

- ✓ A. Survivors of serious acute poisoning may develop delayed neurologic sequelae, especially in the form of Parkinsonian symptoms
- ✓ B. Chronic exposure is associated with headache, vertigo, tremors, weakness, fatigue, confusion, optic neuropathy, seizures, hemiparesis.
- ✓ ***N.B*** foam from the mouth, breathing smells, the vomit also smells

❖ Antidotal therapy:

- The 3-step Eli Lilly cyanide kit approach
 - First step: Amyl *nitrite* every minute until the 2nd step is begun.
 - Second step: Sodium *nitrite* (3% solution) slow IV, i.e. over 5 to 10 minutes.
 - Third step: Sodium thiosulfate

❖ Mechanism of action of nitrites:

- *Nitrites* induce *methemoglobinemia* which causes the detachment of cyanide from the heme group of *cytochrome oxidase*. Amyl nitrite are meant to be a *temporizing* measure until sodium nitrite can be administered intravenously.

• Mechanism of action of sodium thiosulfate:

- It enables the enzyme rhodanese to catabolize cyanide to non-toxic *thiocyanate* which is excreted in the urine.
- rhodanese transfers sulfur from thiosulfate (or another sulfur donor) to CN